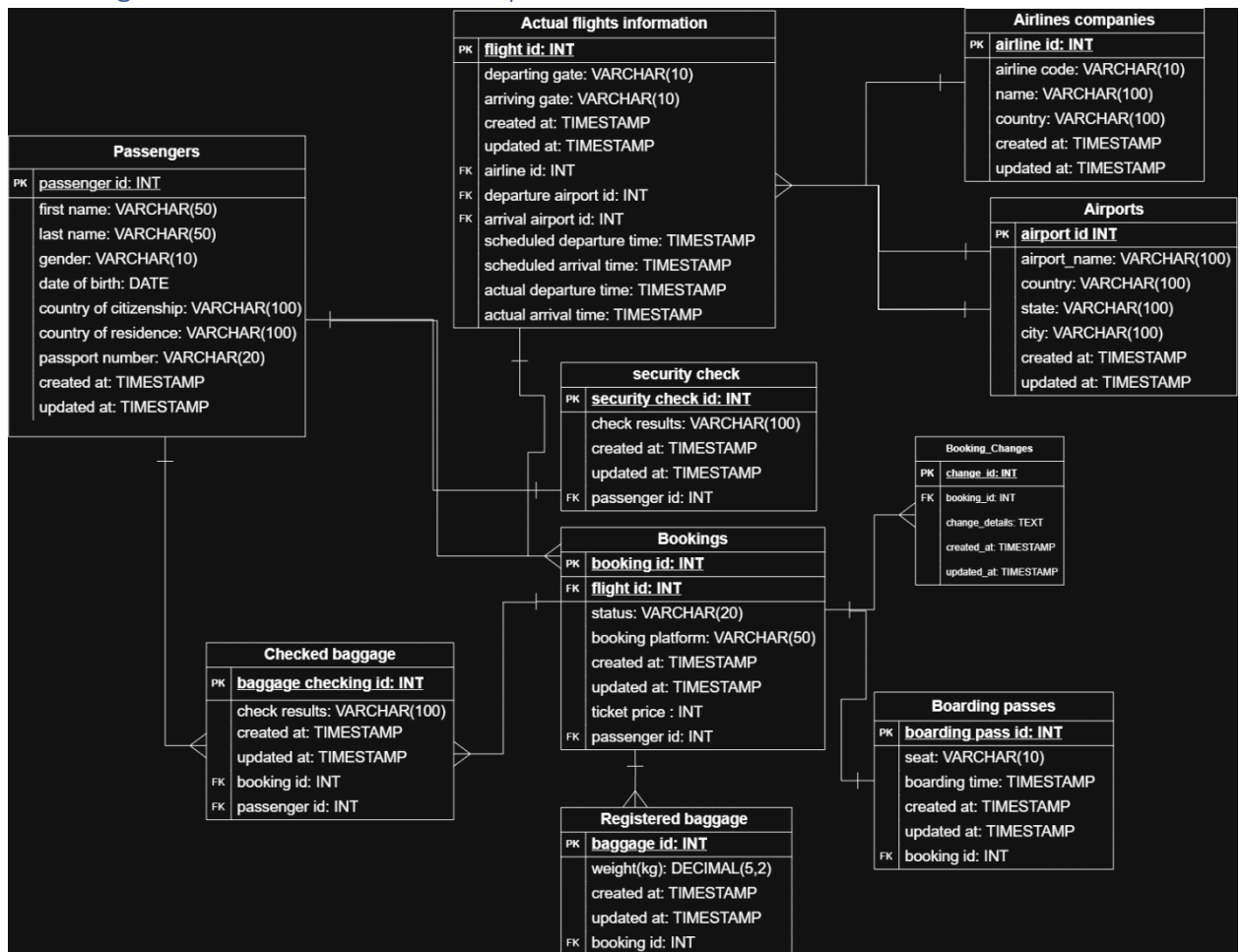


Laboratory Work 1

ERD Diagram for an International Airport Database



1. System Overview

The International Airport Database is designed to manage essential airport operations. The system stores information about airports, airline companies, flights, passengers, bookings, boarding passes, registered baggage, baggage checks, security checks, and booking changes.

The database records both scheduled and actual flight information. Each flight is operated by an airline company and is connected to two airports: one departure airport and one arrival airport. Passengers can create bookings for flights. The system also records boarding passes, registered baggage, baggage-check results, passenger security checks, and changes made to bookings.

2. Entities and Attributes

2.1 Passengers

The `Passengers` entity stores personal information about passengers.

- `passenger_id`: INT — Primary Key
- `first_name`: VARCHAR(50)
- `last_name`: VARCHAR(50)
- `gender`: VARCHAR(10)

- `date_of_birth`: DATE
- `country_of_citizenship`: VARCHAR(100)
- `country_of_residence`: VARCHAR(100)
- `passport_number`: VARCHAR(20)
- `created_at`: TIMESTAMP
- `updated_at`: TIMESTAMP

2.2 Actual Flights Information

The `Actual flights information` entity stores scheduled and actual information about flights.

- `flight_id`: INT — **Primary Key**
- `departing_gate`: VARCHAR(10)
- `arriving_gate`: VARCHAR(10)
- `created_at`: TIMESTAMP
- `updated_at`: TIMESTAMP
- `airline_id`: INT — **Foreign Key**
- `departure_airport_id`: INT — **Foreign Key**
- `arrival_airport_id`: INT — **Foreign Key**
- `scheduled_departure_time`: TIMESTAMP
- `scheduled_arrival_time`: TIMESTAMP
- `actual_departure_time`: TIMESTAMP
- `actual_arrival_time`: TIMESTAMP

2.3 Airlines Companies

The `Airlines companies` entity stores information about airline companies.

- `airline_id`: INT — **Primary Key**
- `airline_code`: VARCHAR(10)
- `name`: VARCHAR(100)
- `country`: VARCHAR(100)
- `created_at`: TIMESTAMP
- `updated_at`: TIMESTAMP

2.4 Airports

The `Airports` entity stores information about airports.

- `airport_id`: INT — **Primary Key**
- `airport_name`: VARCHAR(100)
- `country`: VARCHAR(100)
- `state`: VARCHAR(100)
- `city`: VARCHAR(100)
- `created_at`: TIMESTAMP
- `updated_at`: TIMESTAMP

2.5 Bookings

The `Bookings` entity stores flight reservations made by passengers.

- `booking_id`: INT — **Primary Key**

- `flight_id: INT` — Foreign Key
- `status: VARCHAR(20)`
- `booking_platform: VARCHAR(50)`
- `created_at: TIMESTAMP`
- `updated_at: TIMESTAMP`
- `ticket_price: INT`
- `passenger_id: INT` — Foreign Key

2.6 Booking Changes

The `Booking_Changes` entity stores information about changes made to bookings.

- `change_id: INT` — Primary Key
- `booking_id: INT` — Foreign Key
- `change_details: TEXT`
- `created_at: TIMESTAMP`
- `updated_at: TIMESTAMP`

2.7 Boarding Passes

The `Boarding passes` entity stores boarding passes issued for bookings.

- `boarding_pass_id: INT` — Primary Key
- `seat: VARCHAR(10)`
- `boarding_time: TIMESTAMP`
- `created_at: TIMESTAMP`
- `updated_at: TIMESTAMP`
- `booking_id: INT` — Foreign Key

2.8 Registered Baggage

The `Registered baggage` entity stores baggage registered during a booking.

- `baggage_id: INT` — Primary Key
- `weight_kg: DECIMAL(5,2)`
- `created_at: TIMESTAMP`
- `updated_at: TIMESTAMP`
- `booking_id: INT` — Foreign Key

2.9 Checked Baggage

The `Checked baggage` entity stores baggage-check results at the airport.

- `baggage_checking_id: INT` — Primary Key
- `check_results: VARCHAR(100)`
- `created_at: TIMESTAMP`
- `updated_at: TIMESTAMP`
- `booking_id: INT` — Foreign Key
- `passenger_id: INT` — Foreign Key

2.10 Security Check

The security check entity stores the results of passenger security checks.

- security_check_id: INT — Primary Key
- check_results: VARCHAR(100)
- created_at: TIMESTAMP
- updated_at: TIMESTAMP
- passenger_id: INT — Foreign Key

3. Relationships

The following relationships are represented in the ERD.

1. One airline company can operate many flights. Each flight belongs to one airline company.

Airlines Companies 1 – M Actual Flights Information

2. One airport can be the departure airport for many flights. Each flight has one departure airport.

Airports 1 – M Actual Flights Information (departure airport)

3. One airport can be the arrival airport for many flights. Each flight has one arrival airport.

Airports 1 – M Actual Flights Information (arrival airport)

4. One flight can have many bookings. Each booking belongs to one flight.

Actual Flights Information 1 – M Bookings

5. One passenger can make many bookings. Each booking belongs to one passenger.

Passengers 1 – M Bookings

6. One booking can have many booking changes. Each booking change belongs to one booking.

Bookings 1 – M Booking Changes

7. Each booking is associated with one boarding pass, and each boarding pass belongs to one booking.

Bookings 1 – 1 Boarding Passes

8. One booking can include many registered baggage items. Each baggage item belongs to one booking.

Bookings 1 – M Registered Baggage

9. One booking can have many checked baggage records. Each checked baggage record belongs to one booking.

Bookings 1 – M Checked Baggage

10. One passenger can have many checked baggage records. Each checked baggage record belongs to one passenger.

Passengers 1 — M Checked Baggage

11. Each passenger is associated with a security check. Each security check belongs to one passenger.

Passengers 1 — 1 Security Check

4. Integrity Constraints

Primary keys uniquely identify records in their entities. Foreign keys establish relationships between tables and maintain referential integrity.

The `airline_id` in `Actual flights information` references `Airlines companies`. The `departure_airport_id` and `arrival_airport_id` reference `Airports`. The `flight_id` and `passenger_id` in `Bookings` reference the corresponding flight and passenger records.

The foreign keys in `Booking_Changes`, `Boarding passes`, `Registered baggage`, `Checked baggage`, and `security check` reference their related parent entities. These constraints prevent records from referring to non-existing entities.

5. ERD Notation Key

- **PK** — Primary Key. It uniquely identifies each record in an entity.
- **FK** — Foreign Key. It references the primary key of another entity.
- **1** — One related record.
- **M** — Many related records.
- A single bar in Crow's Foot notation represents one record.
- A crow's foot represents many records.

6. Conclusion

The ERD represents the main processes of an international airport system. It supports the storage and management of airport data, airline companies, flight schedules, passengers, bookings, boarding passes, baggage, baggage checks, security procedures, and booking changes. The defined primary keys, foreign keys, and relationships provide a structured and consistent database design.